

Class 1

The American Economy in 25 Charts

Course overview — the trailer

The American Economy — draft course materials (proposal under discussion)

The point of today's class

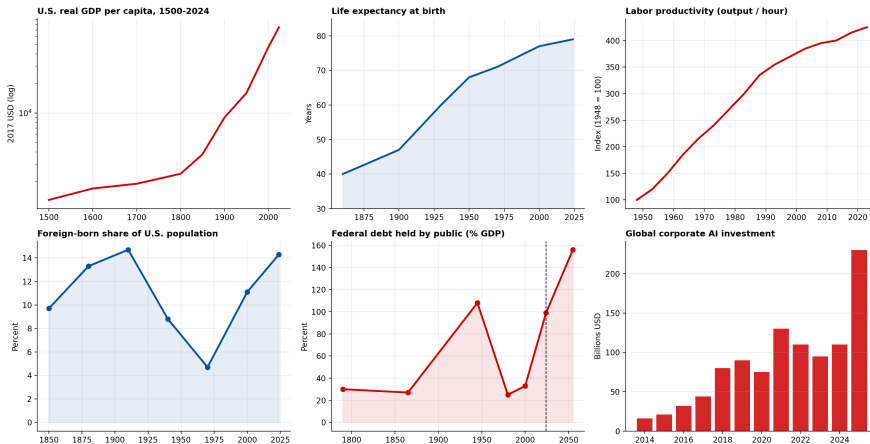
For most of human history, ordinary people were poor, sick, and short-lived. This course explains how Americans became the richest people in history — and asks whether it can last.

The rules of today's class

- ▶ Twenty-five charts go up, one at a time — one for each class this semester.
- ▶ As each one appears, ask yourself the question that runs the whole course: **do I believe this chart?**
- ▶ Every hand that goes up is doing economics: interrogating evidence.
- ▶ By the end of the semester you will know how to attack — and defend — every chart on this screen.

The American economy in six panels

The American Economy in Six Charts



Sources: Maddison Project Database; OWID / CDC; BLS; U.S. Census Bureau; CBO; Stanford HAI AI Index 2023.

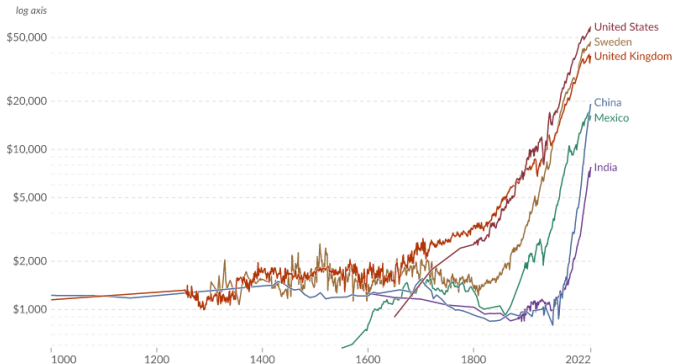
Source: Composite of charts assembled from BEA, BLS, CBO, CDC, Census, EPA, Maddison Project, MPI, NSF, OWID, Peterson-KFF, Stanford HAI.

Class 2 preview — GDP, Living Standards, and Welfare

GDP per capita, 1000 to 2022

Our World
in Data

GDP per capita is a country's gross domestic product divided by its population. This data is adjusted for inflation and differences in living costs between countries.



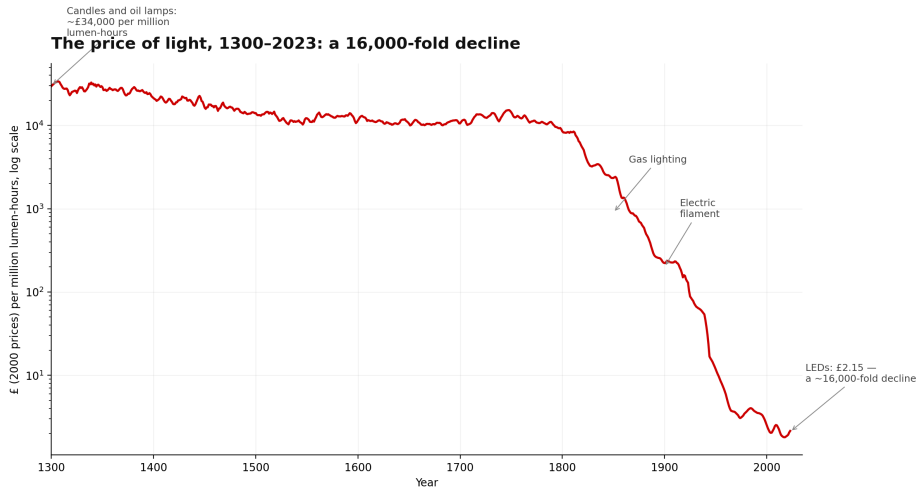
Data source: Bolt and van Zanden – Maddison Project Database 2023

OurWorldinData.org/economic-growth | CC BY

Note: This data is expressed in international-\$ at 2011 prices.

Source: Maddison Project Database 2023, via Our World in Data. CC-BY 4.0.

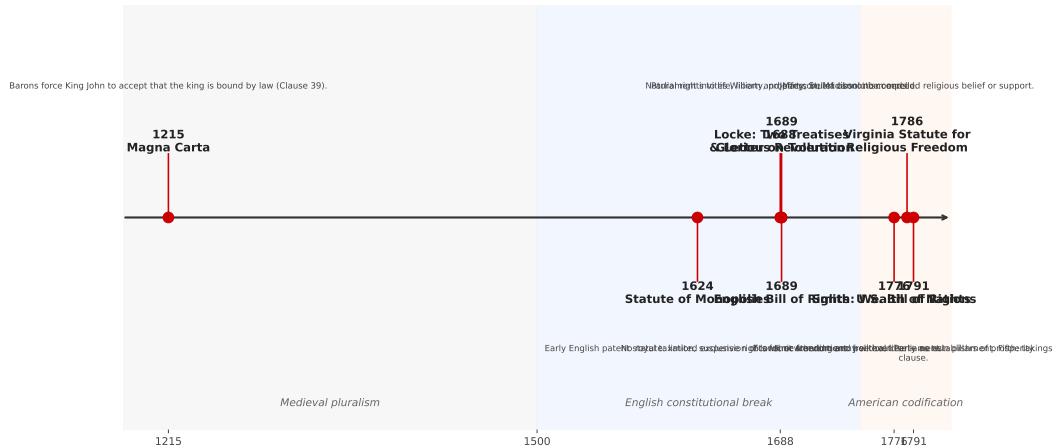
Class 3 preview — The Recipe for Growth



Source: Fouquet and Pearson (2006), updated, via Our World in Data. CC-BY 4.0.

Class 4 preview — Liberty and the Rules of the Game

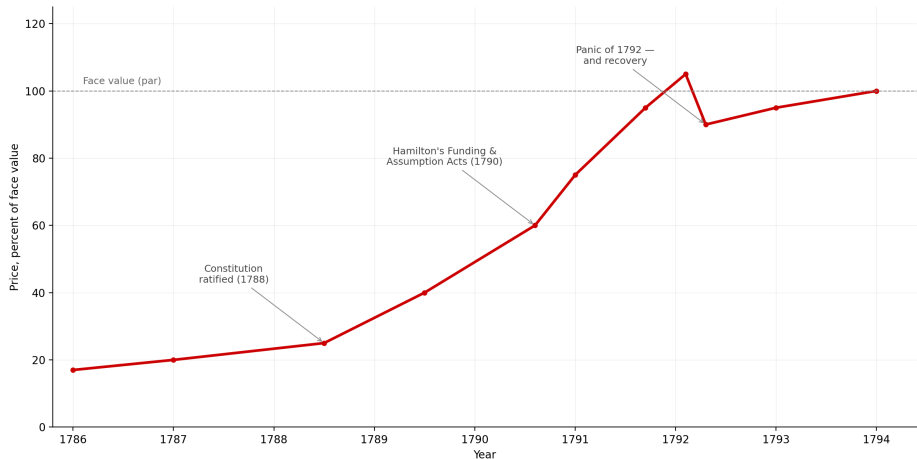
From subject to citizen: the institutional roots of modern prosperity



Source: Author's construction: Magna Carta (1215) to the U.S. Bill of Rights (1791).

Class 5 preview — The Constitution as an Economic Document

The price of American promises: U.S. debt, 1786-1794



Source: Illustrative reconstruction based on Richard Sylla's accounts of founding-era debt prices.

Class 6 preview — Why Nations Diverge



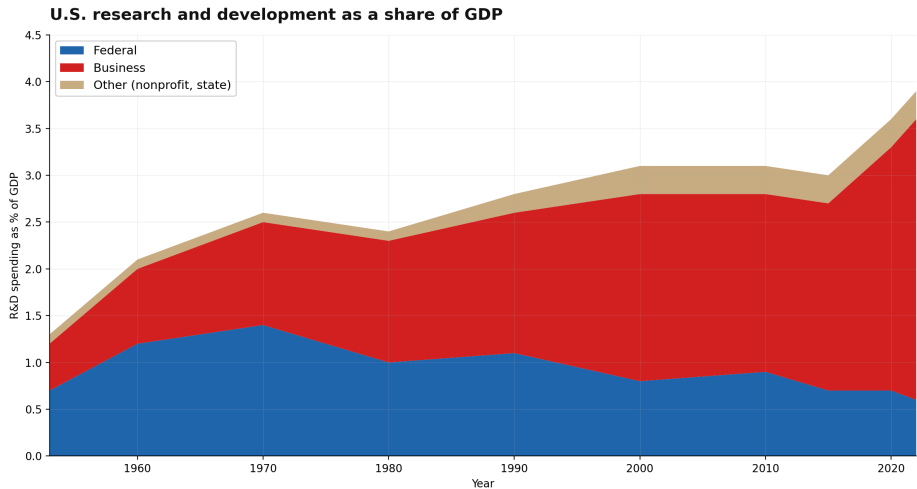
Source: NASA, ISS photograph ISS038-E-038300 (2014). Public domain.

Class 7 preview — Land, Rails, and the National Market



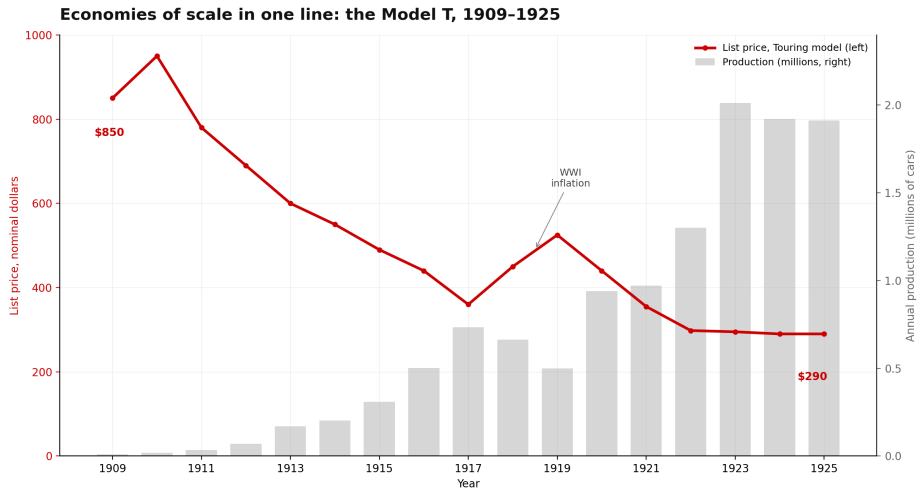
Source: Andrew J. Russell, Promontory Summit, Utah, May 10, 1869. Public domain.

Class 8 preview — The American Invention Machine



Source: National Science Foundation, NCSES.

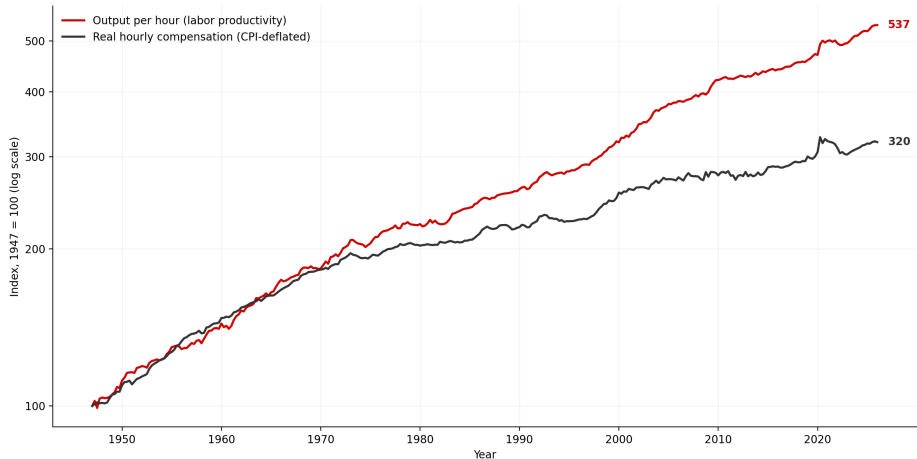
Class 9 preview — Factories, Firms, and Scale



Source: Ford Motor Company data as compiled in Hounshell (1984).

Class 10 preview — Productivity and Wages

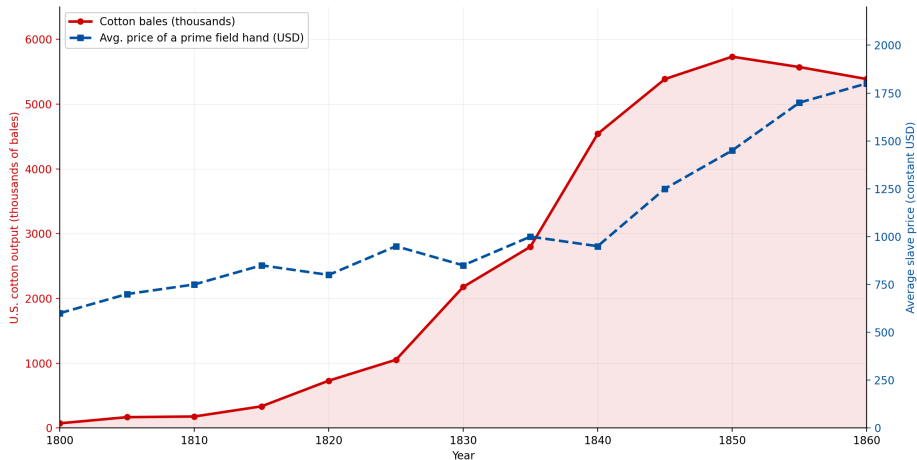
Productivity and real hourly compensation, 1947-present



Source: Bureau of Labor Statistics, via FRED (OPHNFB, COMPRNFB).

Class 11 preview — Free and Coerced Labor

Cotton output and slave prices, antebellum South



Source: Olmstead and Rhode (2008), JEH; Historical Statistics of the United States, Tables Ba4218 and Bb209.

Source: Olmstead and Rhode (2008).

Class 12 preview — Labor Markets and Worker Power

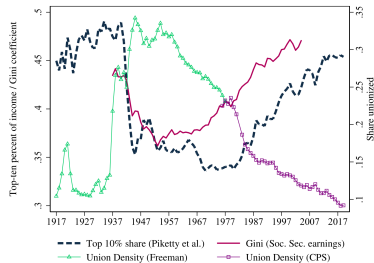


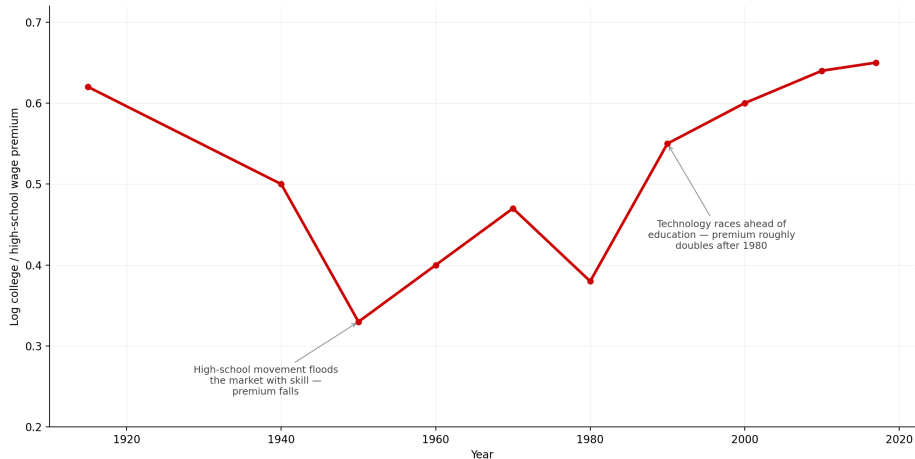
FIGURE I: UNION DENSITY AND INEQUALITY MEASURES, 1917-2019

Data sources: Top share individual income inequality from Piketty, Saez, and Zucman (2018). Union density is number of unionized workers as a share of non-agricultural workforce from Historical Statistics of the United States, together with individual union density as a share of employed civilian workers ages 16 to 65 from the Current Population Survey. We discuss these data sources in detail in Section II.B and Appendix E.

Source: Farber, Herbst, Kuziemko, and Naidu (2021).

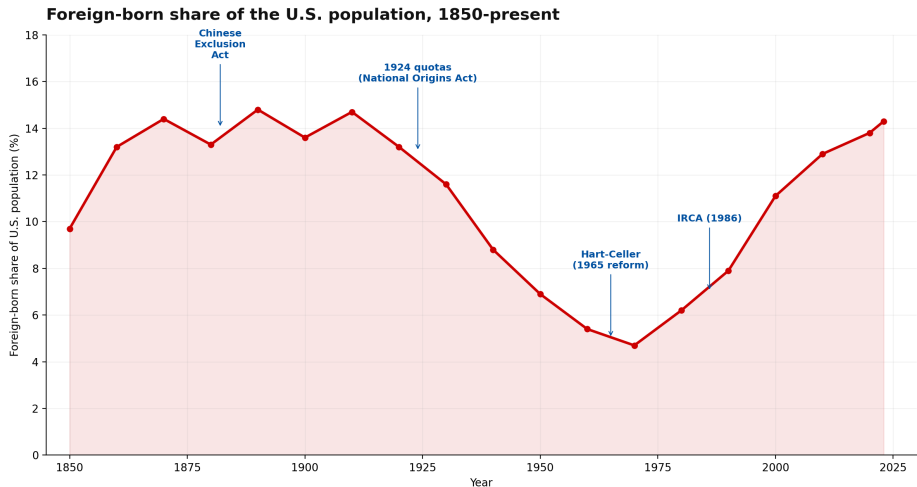
Class 13 preview — Education and Human Capital

The race between education and technology: the college premium, 1915-2017



Source: Reconstructed from Goldin and Katz (2008) and Autor, Goldin, and Katz (2020).

Class 14 preview — Immigration and Assimilation



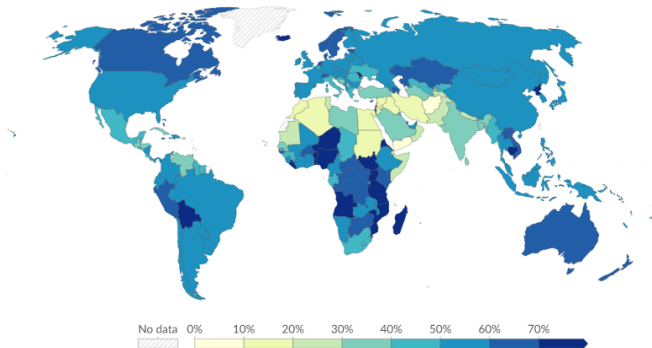
Source: Migration Policy Institute, U.S. Census Bureau data.

Class 15 preview — Women, Work, and Family

Female labor force participation rate, 2025

Share of the female working-age population (ages 15 and over) who are economically active (employed or unemployed).

Our World
in Data



Data source: ILO Modelled Estimates, via World Bank (2026)

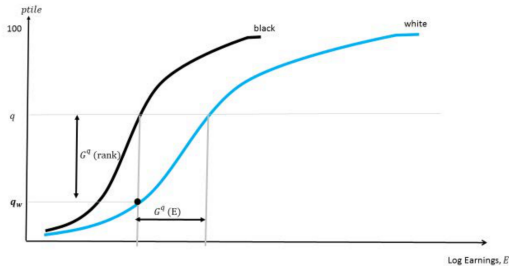
OurWorldinData.org/female-labor-supply | CC BY

Note: All figures correspond to modeled ILO estimates. Employment is defined in line with the 13th ICLS and includes the production of goods for own use, such as subsistence farming.

Source: Goldin (2024), Nobel Lecture.

Class 16 preview — Race, Migration, and Opportunity

Figure 1: Racial Earnings Level and Earning Rank Gaps.

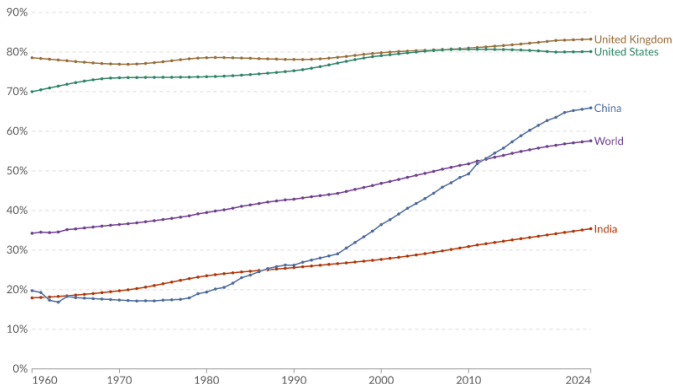


Source: Bayer and Charles (2018).

Class 17 preview — Cities, Housing, and Path Dependence

Share of the population living in urban areas, 1960 to 2024

Our World
in Data



Data source: World Urbanization Prospects - UN Population Division, via World Bank (2026)

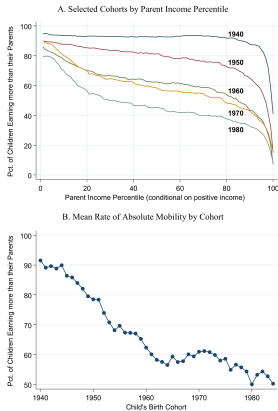
OurWorldinData.org/urbanization | CC BY

Note: Because the estimates of city and metropolitan areas are based on national definitions of what constitutes a city or metropolitan area, cross-country comparisons should be made with caution.

Source: U.S. Census Bureau, Historical Statistics.

Class 18 preview — The American Dream and Mobility

Figure 1. Baseline Estimates of Absolute Mobility by Birth Cohort



Notes: This figure plots the fraction of children earning more than their parents ("absolute mobility") by parental income percentile for selected child birth cohorts (Panel A) and on average by child birth cohort (Panel B). Panel A includes only parents with positive income; within this group, parents' income percentiles are constructed based on their ranks in the distribution of parents' incomes within each child cohort. Panel B includes parents with 0 income. Defines absolute mobility = 100%. For that cohort when computing the mean rate of

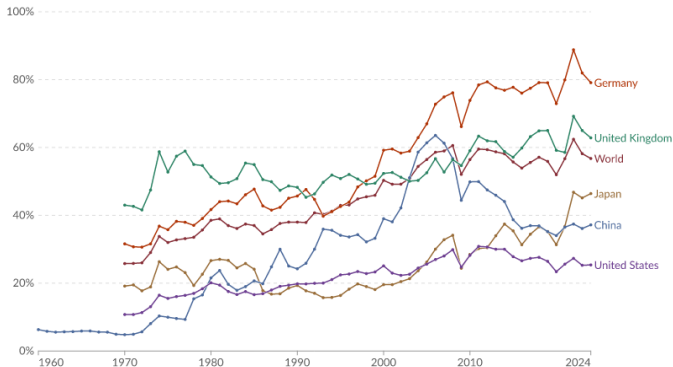
Source: Chetty, Grusky, Hell, Hendren, Manduca, and Narang (2017), Science.

Class 19 preview — Trade, Tariffs, and Globalization

Trade as a share of GDP, 1960 to 2024



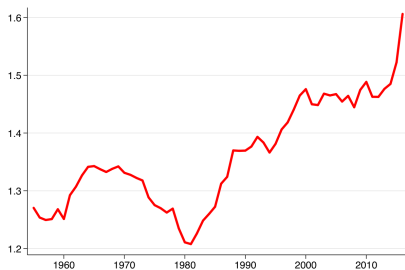
Sum of exports and imports of goods and services, divided by gross domestic product, expressed as a percentage. This is also known as the "trade openness index".



Data source: National statistical organizations and central banks, OECD national accounts, and World Bank staff estimates (2026)
OurWorldinData.org/trade-and-globalization | CC BY

Source: Our World in Data, based on World Bank and historical sources. CC-BY 4.0.

Class 20 preview — Market Power and Antitrust



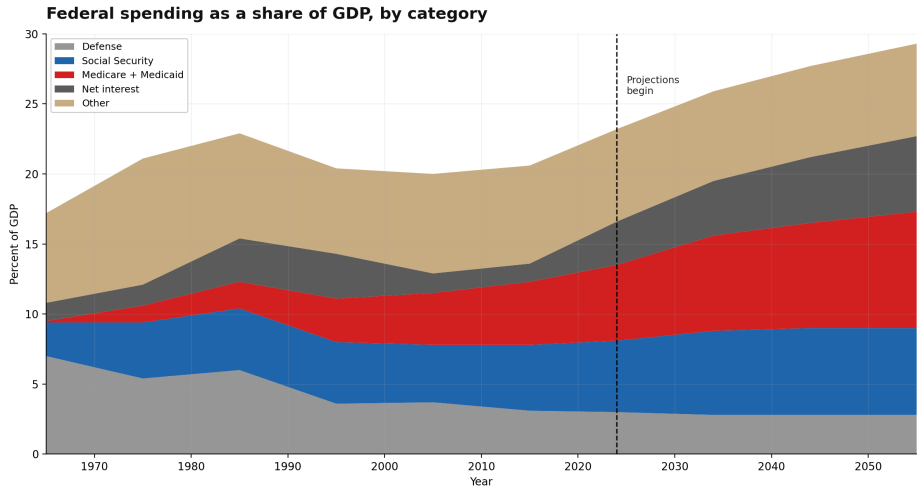
1: Average Markups. Output elasticities θ_{st} from estimated production functions using varying and sector-specific (2 digit). Average is revenue weighted. Evolution 1955-

since 1980 there has been a steady increase to 1.61. In 2016, the average markup is 61% over marginal cost, compared to 21% in 1980. In Online Appendix 5 we report the distribution of individual firms' markups.

In broad terms, there are three sources that can account for this rise in average markups:

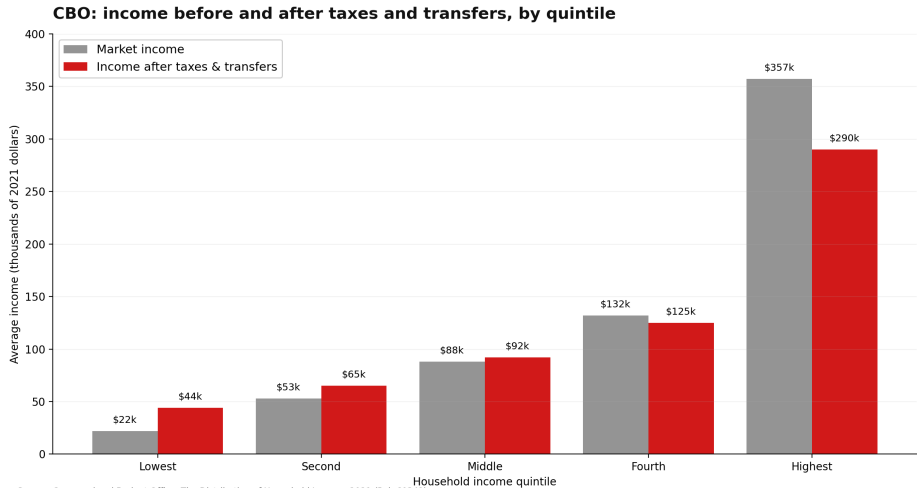
Source: De Loecker, Eeckhout, and Unger (2020).

Class 21 preview — The Useful State



Source: Office of Management and Budget; Congressional Budget Office.

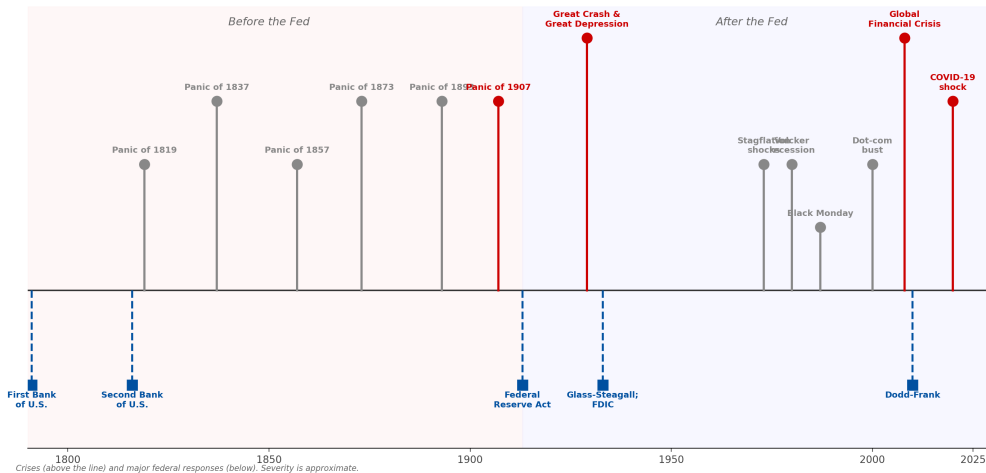
Class 22 preview — Fiscal Capacity



Source: Congressional Budget Office, *The Distribution of Household Income*.

Class 23 preview — Money, Banking, and Crises

U.S. banking and financial crises, and the institutional response



Source: Federal Reserve Bank of New York; Reinhart and Rogoff (2009).

Class 24 preview — Health, Environment, and Externalities

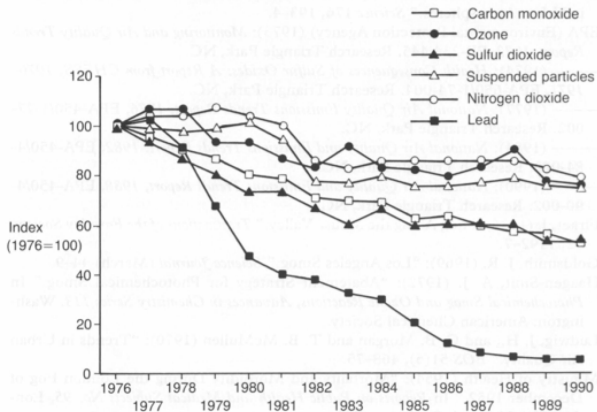
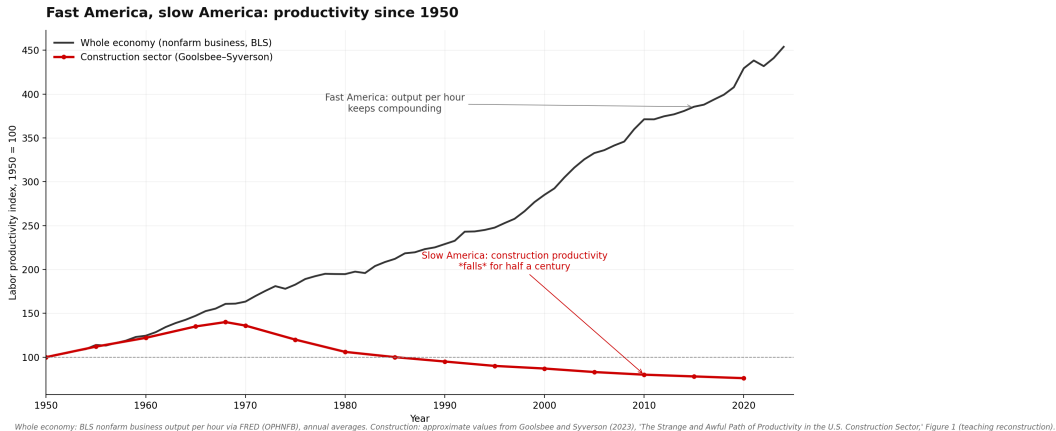


Figure 45.4 Ambient air pollutant concentrations, USA, 1976–90

Source: US Environmental Protection Agency, National Air Quality and Emissions Trends Report, annual.

Source: U.S. Environmental Protection Agency, *Our Nation's Air*.

Class 25 preview — AI and the Institutions of Progress



Source: BLS via FRED; Goolsbee and Syverson (2023), teaching reconstruction.

How this course works

- ▶ **One chart, one idea, one institution per class.** Every lecture is anchored in a chart that demands explanation, an economic concept that explains it, and an American institution that shaped it.
- ▶ **Four acts, one argument:** the growth miracle; building the growth machine; who got access; governing capitalism's consequences.
- ▶ **History paired with the present:** railroads and broadband, the 1924 quotas and the 2021 surge, Standard Oil and Big Tech.
- ▶ **Measurement and causality throughout:** what is the comparison, what is the counterfactual, how could this mislead?

What economists actually do

- ▶ Claudia Goldin reconstructed two centuries of women's work from census manuscripts — and won the Nobel Prize.
- ▶ Raj Chetty used anonymized tax records to map the American Dream, neighborhood by neighborhood.
- ▶ David Autor traced what Chinese import competition did to every American commuting zone.
- ▶ The common toolkit: a hard question, careful data, and an honest comparison. That toolkit is what this course teaches.

Discussion

On your index card: which chart surprised you most — and which one do you *not* believe? (Keep the card. We will return to it in the final class.)

Read before next class

- ▶ The syllabus.
- ▶ Our World in Data, “Economic Growth” — browse the charts before Class 2.